```
/*****************************************************************************************************************
PROGRAM TITLE:      Frailty project documentation - DATA FROM GRDC - AS
ANALYST:            AHMAD SABRA, MS MPH
CREATION DATE:      MARCH 2025
NOTE:               * CODE HAS BEEN ANNOTATED. EXMAPLES AND COMMENTS WERE CREATED FOR ADDITIONAL CONTEXT
                    * USE DATA FROM GRDC (FOR THE RECRUITED MEMBERS)
                    * MERGE BACK WITH THE DATA FROM REDCAP (MERGE BY PATIENT_ID) TO GENERATE A MORE COMPREHENSIVE REPORT
                    * MOST RECENT DATA PULL FILE LOCATION (FEB 2025):
                        \\researchsan02a\Shr3\PM\DC_FK Frailty assessment\RDR FK250117 Frailty

COMMENT:            THIS CODE PULLS DATA FROM MULTIPLE SOURCES:
                        1) SAMPLE
                        2) EXPOSURE
                        3) EMPLOYMENT ON 9/10 (PRE 9/11/2001)
                        4) DEMOGRAPHICS AND EXPOISRE
                        5) MOST RECENT SMOKING STATUS
                        6) MEAN BMI
                        7) CERITIFICAITONS


                    FOR EXAMPLES ON HOW TO REPORT THESE DATA, PLEASE REFER TO OUR PUBLISHED ARTICLE USING
                    DATA FROM THE PILOT STUDY.

                        THOMPSON HM, THANIK E, SABRA A, KO F, HUNG WW, COLICINO E, LUCCHINI RG, BELLO G,
                        CRANE M, TEITELBAUM SL, ORNSTEIN KA. A PILOT STUDY TO IDENTIFY FACTORS ASSOCIATED WITH
                        FRAILTY WITHIN THE WORLD TRADE CENTER GENERAL RESPONDER COHORT.
                        AM J IND MED. 2024 JUL;67(7):582-591. DOI: 10.1002/AJIM.23590.
                        EPUB 2024 MAY 12. PMID: 38735862.
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/*For tbis code, I will be using the data from mosr recent data pull (Feb 2025) to demostrate.*/
/* \\researchsan02a\Shr3\PM\DC_FK Frailty assessment\RDR FK250117 Frailty   */
/*The files are in CSV format, so you would need to import to SAS or R accordingly*/


libname pull "\\researchsan02a\Shr3\PM\DC_FK Frailty assessment\RDR FK250117 Frailty";


/*************************************************************************
I- PREPARE SAMPLE AND CREATE AGE AT MOST RECENT VISIT
*************************************************************************/

/*prepare list of 395 members*/
proc sql;
    create table crosswalk as
    select distinct patient_id
    from pull.Fk250117_demographics
    order by patient_id;
quit;


/*extract current_age (at most recent visit)*/
proc sql;
    create table last as
    select distinct patient_id, max(visit_number) as last_visit, visit_Date as last_visdt
    from pull.Fk250117_visits_marital_work
    group by patient_id
    having visit_number=max(visit_number);

    create table current_age as
    select distinct a.patient_id, a.dob, b.last_visit, intck('year',a.dob,b.last_visdt) as lastVis_Age
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    from pull.Fk250117_demographics a
    left join last b
    on a.patient_id=b.patient_id;
quit;


/* you could also create any other date variables as needed for your analysis.*/
/* For e.g., current_age was requested to be included in the reports*/


/**************************************************************************
2- PULL EXPOSURE DATA FROM EAQ AT VISIT 1
**************************************************************************/


/*PLEASE NOTE, IN FEB 2025, GRDC PROVIDED RECENTLY CREATED EAQ DATA FOR VISIT 2 AND 3, WHICH WERE NEVER USED IN*/
/*ANY OF THE REPORTS OR MANUSCRITPS. THIS IS THE FIRST TIME THIS DATA IS BEING USED. */
/*THE BELOW CODE DOESN'T INCLUDE NEW DATA, RATHER JUST DATA FROM VISIT 1*/


/*Exposure data from EAQ visit 1(asbestos, etc...)*/

%let Asbestos= EAQ_02_06  EAQ_03_06  EAQ_04_06  EAQ_05_06  EAQ_06_06  EAQ_07_06 EAQ_00_06  EAQ_000_06;
%let Cadmium= EAQ_02_08  EAQ_03_08  EAQ_04_08  EAQ_05_08  EAQ_06_08  EAQ_07_08  EAQ_00_08 EAQ_000_08;
%let Diesel= EAQ_02_10 EAQ_03_10 EAQ_04_10 EAQ_05_10 EAQ_06_10 EAQ_07_10 EAQ_00_10 EAQ_000_10;
/*Dust general*/
%let Dustgen= EAQ_02_12 EAQ_03_12 EAQ_04_12 EAQ_05_12 EAQ_06_12 EAQ_07_12 EAQ_00_12 EAQ_000_12;
/*Dust mineral or mining*/
%let Dustmin= EAQ_02_14 EAQ_03_14 EAQ_04_14 EAQ_05_14 EAQ_06_14 EAQ_07_14 EAQ_00_14 EAQ_000_14;
/*Dust silica/sand*/
%let Silica= EAQ_02_16 EAQ_03_16 EAQ_04_16 EAQ_05_16 EAQ_06_16 EAQ_07_16 EAQ_00_16 EAQ_000_16;
/*Dust wood*/
%let Wood= EAQ_02_18 EAQ_03_18 EAQ_04_18 EAQ_05_18 EAQ_06_18 EAQ_07_18 EAQ_00_18 EAQ_000_18;
/*Ergonomic risk factors*/
%let Ergonom= EAQ_02_20 EAQ_03_20 EAQ_04_20 EAQ_05_20 EAQ_06_20 EAQ_07_20 EAQ_00_20 EAQ_000_20;
%let Fiberglass= EAQ_02_22  EAQ_03_22 EAQ_04_22 EAQ_05_22 EAQ_06_22 EAQ_07_22 EAQ_00_22 EAQ_000_22;
/*Heat/Cold Stress*/
%let Heat= EAQ_02_24 EAQ_03_24 EAQ_04_24 EAQ_05_24 EAQ_06_24 EAQ_07_24 EAQ_00_24 EAQ_000_24;
/*Industrial cleaning solutions*/
%let Industry= EAQ_02_26 EAQ_03_26 EAQ_04_26 EAQ_05_26 EAQ_06_26 EAQ_07_26 EAQ_00_26 EAQ_000_26;
%let Lead= EAQ_02_28 EAQ_03_28 EAQ_04_28 EAQ_05_28 EAQ_06_28 EAQ_07_28 EAQ_00_28 EAQ_000_28;
%let Mercury= EAQ_02_30 EAQ_03_30 EAQ_04_30 EAQ_05_30 EAQ_06_30 EAQ_07_30 EAQ_00_30 EAQ_000_30;
%let Noise= EAQ_02_32 EAQ_03_32 EAQ_04_32 EAQ_05_32 EAQ_06_32 EAQ_07_32 EAQ_00_32 EAQ_000_32;
/*Non-diesel exhaust*/
%let Nondiesel= EAQ_02_34 EAQ_03_34 EAQ_04_34 EAQ_05_34 EAQ_06_34 EAQ_07_34 EAQ_00_34 EAQ_000_34;
/*Pesticides*/
%let Pest= EAQ_02_36 EAQ_03_36 EAQ_04_36 EAQ_05_36 EAQ_06_36 EAQ_07_36 EAQ_00_36 EAQ_000_36;
/*Smoke (non-cigarette)*/
%let Noncig= EAQ_02_38 EAQ_03_38 EAQ_04_38 EAQ_05_38 EAQ_06_38 EAQ_07_38 EAQ_00_38 EAQ_000_38;
/*Solvents/gasoline*/
%let Gasoline= EAQ_02_40 EAQ_03_40 EAQ_04_40 EAQ_05_40 EAQ_06_40 EAQ_07_40 EAQ_00_40 EAQ_000_40;
/*Welding fumes or Brazing*/
%let Welding= EAQ_02_42 EAQ_03_42 EAQ_04_42 EAQ_05_42 EAQ_06_42 EAQ_07_42 EAQ_00_42 EAQ_000_42;
/*Other*/
    /*EAQ_02_44 EAQ_03_44 EAQ_04_44 EAQ_05_44 EAQ_06_44 EAQ_07_44 EAQ_00_44 EAQ_000_44(excluded since these are free-txt)0*/



data exposure (drop= &Asbestos &Cadmium &Diesel &Dustgen &Dustmin &Silica &Wood &Ergonom &Fiberglass
            &Heat &Industry &Lead &Mercury &Noise &Nondiesel &Pest &Noncig &Gasoline &Welding);
    set pull.Fk250117_eaq (keep= patient_id
            &Asbestos &Cadmium &Diesel &Dustgen &Dustmin &Silica &Wood &Ergonom &Fiberglass
            &Heat &Industry &Lead &Mercury &Noise &Nondiesel &Pest &Noncig  &Gasoline &Welding) ;
    array Exp1 {8}   &Asbestos;
    array Exp2 {8}   &Cadmium;
    array Exp3 {8}   &Diesel;
    array Exp4 {8}   &Dustgen;
    array Exp5 {8}   &Dustmin;
    array Exp6 {8}   &Silica;
    array Exp7 {8}   &Wood;
    array Exp8 {8}   &Ergonom;
    array Exp9 {8}   &Fiberglass;
    array Exp10 {8}  &Heat;
    array Exp11 {8}  &Industry;
    array Exp12 {8}  &Lead;
    array Exp13 {8}  &Mercury;
    array Exp14 {8}  &Noise;
```

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    array Exp15 {8}  &Nondiesel;
    array Exp16 {8}  &Pest;
    array Exp17 {8}  &Noncig;
    array Exp18 {8}  &Gasoline;
    array Exp19 {8}  &Welding;
        do i = 1 to 8  until (Exp1 {I} =1 | Exp2 {I} =1 | Exp3 {I} =1 | Exp4 {I} =1 | Exp5 {I} =1 |
                                Exp6 {I} =1 | Exp7 {I} =1 | Exp8 {I} =1 | Exp9 {I} =1 | Exp10 {I} =1 |
                            Exp11 {I} =1 | Exp12 {I} =1 | Exp13 {I} =1 | Exp14 {I} =1 | Exp15 {I} =1 |
                                Exp16 {I} =1 | Exp17 {I} =1 | Exp18 {I} =1 | Exp19 {I} =1 );
            if  Exp1{I} = 1 Then Asbestos=1;
            if  Exp2{I} = 1 Then Cadmium=1;
            if  Exp3{I} = 1 Then Diesel=1;
            if  Exp4{I} = 1 Then Dustgen=1;
            if  Exp5{I} = 1 Then Dustmin=1; ;
            if  Exp6{I} = 1 Then Silica=1;
            if  Exp7{I} = 1 Then Wood=1;
            if  Exp8{I} = 1 Then Ergonom=1;
            if  Exp9{I} = 1 Then Fiberglass=1;
            if  Exp10{I} = 1 Then Heat=1;
            if  Exp11{I} = 1 Then Industry=1;
            if  Exp12{I} = 1 Then Lead=1;
            if  Exp13{I} = 1 Then Mercury=1;
            if  Exp14{I} = 1 Then Noise=1;
            if  Exp15{I} = 1 Then Nondiesel=1;
            if  Exp16{I} = 1 Then Pest=1;
            if  Exp17{I} = 1 Then Noncig=1;
            if  Exp18{I} = 1 Then Gasoline=1;
            if  Exp19{I} = 1 Then Welding=1;
        end;
    drop i;

    ARRAY MISS {19}     Asbestos Cadmium Diesel Dustgen Dustmin Silica Wood Ergonom Fiberglass
                        Heat Industry Lead Mercury Noise Nondiesel Pest Noncig Gasoline Welding;
    do i = 1 to 19;
        if MISS {i}= . then MISS {i}=0;
    end;

run;


/***************************************************************************
3- PULL EMPLOYMENT ON 9/10 (PRE 9/11/2001) DATA
***************************************************************************/


/*merge back with 'Frailty_all', create new variables 'pre9110ccup & 'pre911_group' using data from 'EAQ_01_04'*/

data employment_pre911;
    SET pull.Fk250117_eaq (keep=subject_id EAQ_01_04 in=b);
    length pre9110ccup $100.;
    select (EAQ_01_04);
        when ('DirectorofBuildingandCodeEnforcement,TownofQueensbury', 'ExecutivePresident', 'SecruityManager', 'accountmanag
            'executiveasst.', 'manager', 'supervisor park department')
                pre9110ccup= '11: Management Occupations';
        when ('Computer Security') pre9110ccup= '15: Computer & Mathematical Occupations';
        when ('DOTURBANPLANNER', 'EngineerfortheBoardofEducation')
                pre9110ccup=    '17: Architecture and Mathematical Occupations';
        when ('SciencetistArchaeology', 'Research Scientist')
                pre9110ccup= '19: Life, Physical, and Social Science Occupations';
        when ('NYStatecourtofficer', 'courtofficer')
                pre9110ccup= '23: Legal Occupations';
        when ('Artist','BroadcastEngineer','Producer/Director','consultant/audioengineer')
                pre9110ccup= '27: Arts, Design, Entertainment, Sports & Media Occupations';
        when ('PARAMEDIC', 'registerednurse')
                pre9110ccup= '31: Healthcare Support Occupations';
        when ('Commander-centralparkprecinct', 'DeputyInspector-NYPD', 'DeputySheriff', 'Detective',
                'DetectiveforNYPD', 'Firefighter/asstchief', 'MTAPOLICEDEPARTMENT','N.Y.P.D-PoliceOfficer',
                'NYPD', 'NYPD DETECTIVE', 'NYPDPoliceOfficer', 'NYPDTraffic/TowTruckOperator', 'NYPoliceSergeant',
                'PENNSYLVANIA STATE CONSTABLE', 'POLICE OFFICER', 'Police','Police Officer', 'PoliceOfficer',
                'PoliceOfficerLT.', 'PoliceSgt.', 'SargentNYPD', 'Sergaent', 'Sergeant', "Sergeant with Queen's DA office",
                'captainnypd', 'detective', 'firefighter', 'newyorkcitydetective', 'nypd', 'nys zone seargent',
                'policeofficer', 'sargent', 'sergeant', 'state of ny fire saftey', 'state trooper',
                'NYCPoliceOfficer', 'PoliceOfficer', 'nypd','policeofficer')
                pre9110ccup= '33: Protective Service Occupations';
        when ('MAINTENANCEFEDERALRESERVEBANK', 'sanitationworker', 'windowcleaner')
                pre9110ccup= '37: Building and Grounds Cleaning and Maintenance Occupations';
        when ('CentralOffice', 'ClericalAssociate', 'Procurement-OfficeofChiefMedicalExaminer')
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                pre911Occup= '43: Office and Administrative Support Occupations';
        when ('Enviromentalprogramspecialist','envirmental conservation officer','TimberMan')
                pre911Occup= '45: Farming, Fishing, and Forestry Occupations';
        when ('Carpenter', 'Mason', 'Plumber', 'TowOperator', 'Bricklayer', 'contruction officer', 'ironworker')
                pre911Occup= '47: Construction Occupations';
        when ('Field Technician for Verizon', 'electrician', 'fed tech', 'lighttech' , 'Electrician', 'fieldtech')
                pre911Occup= '49: Electrical, Telecommunications & Other Installation & Repair';
        when ('3rdrailoperations(nycta)', 'NYCTransitmachinist', 'TBTABRIDGESANDTUNNELS' , 'Busoperator')
                pre911Occup= '53: Transportation and Material Movers';
        when ('In the Army', 'NYS TROOPER')
                pre911Occup= '55: Military Specific Occupations';
        when ('retired', 'Retired')
                pre911Occup= 'RETIRED';
        when ('unemployed')
                pre911Occup= 'UNEMPLOYED';
        when ('same')
                pre911Occup= 'UNKNOWN';
    otherwise;
    end;

/*  categorize into smaller groups*/
    length pre911_group $100.;
    select (pre911Occup);
        when ('33: Protective Service Occupations', '55: Military Specific Occupations' ) pre911_group= 'Protective Services/I
        when ('47: Construction Occupations' ) pre911_group= 'Construction';
        when ('49: Electrical, Telecommunications & Other Installation & Repair' ) pre911_group= 'Electrical, Telecom, Other
        when ('53: Transportation and Material Movers' ) pre911_group= 'Transportation and Material Moving';
        when ('11: Management Occupations' ) pre911_group= 'Management';
        when ('UNEMPLOYED' ) pre911_group= 'Unemployed/Retired';
        when ('UNKNOWN' ) pre911_group= 'Not Reported';
    otherwise  pre911_group= 'Other';
    end;

run;


/*************************************************************************
4- ADD EXPOSURE DATA AND PULL OTHER DEMOGRAPHIC DATA
*************************************************************************/

proc sql;
    create table demo as
    select distinct a.patient_id
                , a.age911, b.lastVis_Age, intck('year',a.dob,'02MAR2025'd) as age_asof250302
                , a.gender, a.RaceHispanic, a.educ, hispanic, race, race4, racecombo, racepft
                , d.exp4_Imputed
                , ArrivalDust, Arrival, TOTALMONTHSONSITE
                , Asbestos ,Cadmium ,Diesel ,Dustgen ,Dustmin ,Silica ,Wood ,Ergonom ,Fiberglass
                , Heat ,Industry ,Lead ,Mercury ,Noise ,Nondiesel ,Pest ,Noncig ,Gasoline ,Welding
        from pull.Fk250117_demographics a
        left join current_age b
            on a.patient_id=b.patient_id
        left join pull.Fk250117_eaq d
            on a.patient_id=d.patient_id
        left join exposure e
            on a.patient_id=e.patient_id;
quit;

*create yes/no format;
PROC FORMAT;
   VALUE yesnofmt
       0    = 'No'
       1    = 'Yes';
RUN;

*create age groups;
data demo_agegrps;
    format age50_asof250302 yesnofmt.;
    set demo;

    select;
            when (      age911        <=.z)  age911_grp        = 'Unknown';
            when (.z <  age911        < 18)  age911_grp        = '< 18';
            when (18 <= age911        < 25)  age911_grp        = '18-24';
            when (25 <= age911        < 35)  age911_grp        = '25-34';
            when (35 <= age911        < 45)  age911_grp        = '35-44';
            when (45 <= age911        < 55)  age911_grp        = '45-54';
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            when (55 <= age911        < 65)  age911_grp        = '55-64';
            when (65 <= age911        < 75)  age911_grp        = '65-74';
            when (75 <= age911           )  age911_grp        = '75+';
        end;

    select;
            when (      age_asof250302 <=.z)  age_asof250302_grp = 'Unknown';
            when (18 <= age_asof250302 < 45)  age_asof250302_grp = '18-44';
            when (45 <= age_asof250302 < 55)  age_asof250302_grp = '45-54';
            when (55 <= age_asof250302 < 65)  age_asof250302_grp = '55-64';
            when (65 <= age_asof250302 < 75)  age_asof250302_grp = '65-74';
            when (75 <= age_asof250302     )  age_asof250302_grp = '75+';
        end;

    select;
            when (      lastVis_Age <=.z)  lastVis_Age_Group = 'Unknown';
            when (.z <  lastVis_Age < 18)  lastVis_Age_Group = '< 18';
            when (18 <= lastVis_Age < 25)  lastVis_Age_Group = '18-24';
            when (25 <= lastVis_Age < 35)  lastVis_Age_Group = '25-34';
            when (35 <= lastVis_Age < 45)  lastVis_Age_Group = '35-44';
            when (45 <= lastVis_Age < 55)  lastVis_Age_Group = '45-54';
            when (55 <= lastVis_Age < 65)  lastVis_Age_Group = '55-64';
            when (65 <= lastVis_Age < 75)  lastVis_Age_Group = '65-74';
            when (75 <= lastVis_Age     )  lastVis_Age_Group = '75+';
        end;

        format age_asof250302 yesnofmt.;
        *create age variable that represents members over 50 years old as of 03/02/2025;
        if age_asof250302 GE 50 then age50_asof250302=1;
            else age50_asof250302=0;

        if ArrivalDust in (.,.M,0) then ArrivalDust=.M;
        if Arrival in (.,.M,0) then Arrival=.M;
        if exp4_Imputed in (., .M,4,5) then exp4_Imputed=.M;
        if TotalMonthsOnSite in (., .M) then TotalMonthsOnSite=.M;
        if educ in (.,.M,4) then educ=4;

        /*  categorize dust vs no dust*/
    length Dustcat $7.;
    if ArrivalDust=1 then Dustcat="Dust"; else Dustcat="No Dust";
run;


/**************************************************************************
5- PULL SMOKING STATUS DATA FROM THE MULTIPLE IAMQ DATASETS
**************************************************************************/

proc sql;
    create table smoke as
    select PATIENT_ID, visit_number, smoking_status  from pull.Fk250117_iamqv1tdb
        outer union corr
    select PATIENT_ID, visit_number, smoking_status  from pull.Fk250117_iamqv2on
    order by PATIENT_ID,  visit_number;
quit;

data smokex;
    merge crosswalk (in=a) smoke ;
    by first_name last_name ;
    if a;
    if visit_number=. then delete;
run;


*remove duplicates;
proc sort data=smokex out=smoke_nodups nodupkey; by patient_id visit_number;run;
/*no dups*/

*keep only recent 'non-missing' smoking status (i.e., at last visit). ;
proc sql;
    create table recent_smoke as
    select patient_id, max(visit_number) as last_visit, smoking_status as lastvisit_smoke label="Smoking status at last visit
    from smokex
    group by patient_id
    having visit_number=max(visit_number) ;
quit;

data recent_smoke;
```

```
    set recent_smoke;
    if patient_id in (90,99) then lastvisit_smoke='Never smoker';
run;


/************************************************************************
6- PULL MEAN BMI FROM PE DATASET (FOR ALL VISITS)
************************************************************************/

*pull BMI at all visits;

proc sql;
    create table PE as
    select patient_id,  visit_number, height_in, weight_lb, bmi_lbin
            from pull.Fk250117_pe
    order by patient_id, visit_number;
quit;


data PEx;
    merge crosswalk (in=a) PE;
    by first_name last_name ;
    if a;
run;

proc sort data=PEx nodupkey; by patient_id  visit_number ; run;


%let heightLower =  53;
%let heightUpper =  84;
%let weightLower =  75;
%let weightUpper = 600;

data bmi;
    format patient_id visit_Num weight_lb height_in bmi_lbin;
    set PEx(in = p keep = patient_id visit_Number height_in weight_lb bmi_lbin rename=(visit_Number=visit_Num));

    if height_in = 999 or height_in <= 0
        then height_in = .;
    if weight_lb = 999 or weight_lb <= 0
        then weight_lb = .;

    if weight_lb > .z
        then weight_lb = round(weight_lb, 0.1); * No scale measures to hundredths of a pound. *;

    * bmi_lbin is not actually in inches and pounds, but kilograms and meters. *;
    label bmi_lbin  = 'BMI (kg/m�)'
          visit_Num = 'Visit number';
    rename bmi_lbin = bmi_kgm;

    height_orig = height_in;
    weight_orig = weight_lb;

    * Remove extreme impossible values. *;
    if not (&heightLower < height_in < &heightUpper)
        then height_in = .;
    if not (&weightLower < weight_lb < &weightUpper)
        then weight_lb = .;
run;


data BMIx (compress = binary);
    format patient_id visit_Num weight_lb weight_orig height_in height_orig bmi_kgm;
    set BMI (keep = patient_id visit_Num weight_lb weight_orig height_in height_orig bmi_kgm);

    select;
        when (weight_orig <= .z or height_orig <= .z) do;
            bmi_kgm  = .;
            bmi_kgmR = .;
        end;
        when (weight_lb >  .z and height_in >  .z) do;
            bmi_kgm  =       (weight_lb  /2.204622621848776)/((height_in  *0.0254)**2);
            bmi_kgmR = round((weight_lb  /2.204622621848776)/((height_in  *0.0254)**2), 0.1);
        end;
        when (weight_lb <= .z and height_in >  .z) do;
            bmi_kgm  =       (weight_orig/2.204622621848776)/((height_in  *0.0254)**2);
            bmi_kgmR = .;
```

```
        end;
        when (weight_lb >  .z and height_in <= .z) do;
            bmi_kgm  =      (weight_lb  /2.204622621848776)/((height_orig*0.0254)**2);
            bmi_kgmR = .;
        end;
        when (weight_lb <= .z and height_in <= .z) do;
            bmi_kgm  =      (weight_orig/2.204622621848776)/((height_orig*0.0254)**2);
            bmi_kgmR = .;
        end;
    end;
run;

proc sql;
    title "do mean of BMI to include some uncertainy";
    create table bmixx as
    select distinct patient_id,  round ( (mean (bmi_kgm)), 0.1) as BMI_mean
    from bmix
    group by patient_id
    order by patient_id;
quit;


data BMI_mean;
    set BMIxx;
    * BMI - recoding to CDC cutoff and checking for '999' BMI values (recode as missing);

    format bmi_Group $15.;
    select;
        when (         BMI_mean <  0   )  bmi_Group = 'Not reported';
        when ( 0    <  BMI_mean < 18.5)  bmi_Group = 'Underweight';
        when (18.5 <= BMI_mean < 25.0)  bmi_Group = 'Normal';
        when (25.0 <= BMI_mean < 30.0)  bmi_Group = 'Overweight';
        when (30.0 <= BMI_mean        )  bmi_Group = 'Obese';
    end;
run;


/**************************************************************************
7- PULL CERTIFICTIONS DATA
**************************************************************************/

proc sql;
    select comment, count (distinct patient_id) as participants
    from PULL.FK250117_CERTIFICATIONS
    group by comment;
quit;

/*101 MEMBERS had "No certification data available for this member", i.e. are not certified for WTC-related conditions  */

/*import this file "All_Related_180403_ICD9to10.xlsx" to catrgorize certifications*/

proc import datafile="DA\Ahmad\Aging and Fragility\All_Related_180403_ICD9to10.xlsx"
    out= All_Related_180403_ICD9to10
    dbms=xlsx replace;
    getnames = yes;
run;



/*create macro for certifications*/
%macro certs (dsn1, dsn2);


proc sql;
    create table &dsn1._certs as
    select *
    from PULL.FK250117_CERTIFICATIONS
    order by patient_id;

    create table &dsn1._Certs1 as
    select *
    from &dsn1._certs a
    left join   All_related_180403_icd9to10 (keep=code code_description care_suite_detail) b
    on a.Related_Condition=b.code
    order by a.patient_id, a.Start_Date, a.Related_Condition, a.Associated_Condition;
quit;
```

```
/*Recoding the certifications*/

data &dsn1._Certs2;
    set &dsn1._Certs1;
    format CertCat $25.;
    if Certification_Category = "ADJ"  then CertCat = "ADJ";
    if Certification_Category = "ANX" then CertCat = "ANX";
    if Certification_Category = "Cancer"  then CertCat = "Cancer";
    if Certification_Category = "DEP"  then CertCat = "DEP";
    if Certification_Category = "EXT" then CertCat = "EXT";
    if Certification_Category = "GERD"  then CertCat = "GERD";
    if Certification_Category = "HD" then CertCat = "HT";
    if Certification_Category = "ILD"  then CertCat = "ILD";
    if Certification_Category = "OAD"  then CertCat = "OAD";
    if Certification_Category = "PTSD"  then CertCat = "PTSD";
    if Certification_Category = "SA"  then CertCat = "SA";
    if Certification_Category = "SP" then CertCat = "SP";
    if Certification_Category = "SRC" then CertCat = "SRC";
    if Certification_Category = "URD" then CertCat = "URD";
    if Episode_Type not in ("" "WTC-2" "WTC-2 Certification" "WTC-3" "WTC-3 Certification") then CertCat = "WTC1";
    if Related_Condition = "" then CertCat = "None";
run;

/*options nodate pageno = 1;*/
/*PROC FREQ DATA=Certs2 ORDER=FORMATTED;*/
/*  TABLES Certification_Category Care_Suite_Detail ;*/
/*RUN;*/

/*create those with no certification*/

proc sql;
    create table &dsn1._NoCertCategory as
    select distinct *
    from &dsn1._Certs2
    where Related_Condition = '' and CertCat = ''
    order by Related_Condition;
quit;

/*get the codes for conditions and the new certification category*/

options nodate nocenter pageno = 1;
proc sql;
    select distinct compress('"'||Related_Condition||'"') as RC
    from &dsn1._Certs2
    where Related_Condition ne "" and CertCat ne ""
    order by Related_Condition;
quit;


/*Recode certification categories*/

data &dsn1._Certs3;
    set &dsn1._Certs2;
    if related_condition in ("135.0" "D86.0" ) and Certification_Category = "" then CertCat = "SRC";
    if related_condition in ("160" "182" "200" "239" "172.9" "173.9" "C43.9" "C44.90" "C44.91" "C61" ) and Certification_Cate
    if related_condition in ("300" "300.2") and Certification_Category = "" then CertCat = "ANX";
    if related_condition in ("309" "309.90") and Certification_Category = "" then CertCat = "ADJ";
    if related_condition in ("311.0" "296.3" "296.30" "300.4" ) and Certification_Category = "" then CertCat = "DEP";
    if related_condition in ("309.81" "F43.10" ) and Certification_Category = "" then CertCat ="PTSD";
    if related_condition in ("354") and Certification_Category = "" then CertCat = "EXT";
    if related_condition in ("472.00" "472" "472.20" "472.2" "474" "476"  "476.0" "476.00" "477.0" "478" "V58.74" "473.0" "J3
    if related_condition in ("491" "491.0" "491.2" "491.20" "493.90"  "493" "496" "496.0" "496.00" "506.4" "506.40") and Cert
    if related_condition in ("530.81" "535" "535.0" "535.5" "K21.9") and Certification_Category = "" then CertCat = "GERD";
    if related_condition in ("724") and Certification_Category = "" then CertCat = "SP";
    if related_condition in ("302.74" "308.9" "466.19") then CertCat = "Other";
    if related_condition in ("389.0" "517.8" "729.1") or (related_condition = "" and CertCat = "") then CertCat = "None";

    if related_condition = "135.0" and Certification_Category = "" then CertCat = "SRC";
    if related_condition in ("160" "200") and Certification_Category = "" then CertCat = "Cancer";
    if related_condition in ("300") and Certification_Category = "" then CertCat = "ANX";
    if related_condition in ("309") and Certification_Category = "" then CertCat = "ADJ";
    if related_condition in ("311.0") and Certification_Category = "" then CertCat = "DEP";
    if related_condition in ("354") and Certification_Category = "" then CertCat = "EXT";
    if related_condition in ("472.00" "474" "478") and Certification_Category = "" then CertCat = "URD";
    if related_condition in ("493" "496.0" "496.00") and Certification_Category = "" then CertCat = "OAD";
    if related_condition in ("515.0" "516") and Certification_Category = "" then CertCat = "ILD";
```

```
    if related_condition in ("724") and Certification_Category = "" then CertCat = "SP";
    if related_condition in ("308.9" ) then CertCat = "Other";
    if related_condition in ("517.8"  "231" "473.90" "477.1" "518.89" "726" "786.2") or (related_condition = "" and CertCat =

    if related_condition in ("300") and Certification_Category = "" then CertCat = "ANX";
    if related_condition in ("311.0") and Certification_Category = "" then CertCat = "DEP";
    if related_condition in ("354") and Certification_Category = "" then CertCat = "EXT";
    if related_condition in ("472.00" "474" "478") and Certification_Category = "" then CertCat = "URD";
    if related_condition in ("493" "496.0") and Certification_Category = "" then CertCat = "OAD";
    if related_condition in ("516") and Certification_Category = "" then CertCat = "ILD";

    if related_condition in ("200") and Certification_Category = "" then CertCat = "Cancer";
    if related_condition in ("300") and Certification_Category = "" then CertCat = "ANX";
    if related_condition in ("311.0") and Certification_Category = "" then CertCat = "DEP";
    if related_condition in ("354") and Certification_Category = "" then CertCat = "EXT";
    if related_condition in ("474" "478") and Certification_Category = "" then CertCat = "URD";
    if related_condition in ("493" "496.0" "506.40") and Certification_Category = "" then CertCat = "OAD";
    if related_condition = "135.0" and Certification_Category = "" then CertCat = "SRC";
    if related_condition in ("231") and Certification_Category = "" then CertCat = "Cancer";
    if related_condition in ("300") and Certification_Category = "" then CertCat = "ANX";
    if related_condition in ("309") and Certification_Category = "" then CertCat = "ADJ";
    if related_condition in ("311.0") and Certification_Category = "" then CertCat = "DEP";
    if related_condition in ("354") and Certification_Category = "" then CertCat = "EXT";
    if related_condition in ("474" "477.0" "478") and Certification_Category = "" then CertCat = "URD";
    if related_condition in ("493" "496.0")and Certification_Category = "" then CertCat = "OAD";
    if related_condition in ("515.0") and Certification_Category = "" then CertCat = "ILD";
    if related_condition in ("535.5") and Certification_Category = "" then CertCat = "GERD";
    if related_condition in ("724" "724.2") and Certification_Category = "" then CertCat = "SP";
    if related_condition in ("517.8") then CertCat = "Other";
    if related_condition in ("209.3") or (related_condition = "" and CertCat = "") then CertCat = "None";

    if related_condition in ("311.0") and Certification_Category = "" then CertCat = "DEP";
    if related_condition in ("354") and Certification_Category = "" then CertCat = "EXT";
    if related_condition in ("493")and Certification_Category = "" then CertCat = "OAD";
    if related_condition in ("238") or (related_condition = "" and CertCat = "") then CertCat = "None";

    if related_condition in ("200") and Certification_Category = "" then CertCat = "Cancer";
    if related_condition in ("300") and Certification_Category = "" then CertCat = "ANX";
    if related_condition in ("311.0") and Certification_Category = "" then CertCat = "DEP";
    if related_condition in ("493" "496.0") and Certification_Category = "" then CertCat = "OAD";

run;

/*proc freq data=Certs3;*/
/*tables CertCat;*/
/*run;*/


/*Selecting distinct members by certification category (care suites)*/

/*Cancer*/
proc sql;
    create table &dsn1._cancer_patient_ids as
    select distinct patient_id, 1 as Cancer
    from &dsn1._Certs3
    where  CertCat = "Cancer"
    order by patient_id;

/*Adjustment*/
    create table &dsn1._Adj_patient_ids as
    select distinct patient_id, 1 as Adjustment
    from &dsn1._Certs3
    where  CertCat = "ADJ"
    order by patient_id;

/*Anxiety*/
    create table &dsn1._Anx_patient_ids as
    select distinct patient_id, 1 as Anxiety
    from &dsn1._Certs3
    where  CertCat = "ANX"
    order by patient_id;

/*Depression*/
    create table &dsn1._Dep_patient_ids as
    select distinct patient_id, 1 as Depression
    from &dsn1._Certs3
    where  CertCat = "DEP"
```

```
    order by patient_id;

/*Extremities*/
    create table &dsn1._Ext_patient_ids as
    select distinct patient_id, 1 as Extremities
    from &dsn1._Certs3
    where  CertCat = "EXT"
    order by patient_id;

/*GERD*/
    create table &dsn1._GERD_patient_ids as
    select distinct patient_id, 1 as GERD
    from &dsn1._Certs3
    where  CertCat = "GERD"
    order by patient_id;

/*Head*/
    create table &dsn1._Head_patient_ids as
    select distinct patient_id, 1 as Head
    from &dsn1._Certs3
    where  CertCat = "HT"
    order by patient_id;

/*ILD*/
    create table &dsn1._ILD_patient_ids as
    select distinct patient_id, 1 as IntLungDis
    from &dsn1._Certs3
    where  CertCat = "ILD"
    order by patient_id;

/*OAD*/
    create table &dsn1._OAD_patient_ids as
    select distinct patient_id, 1 as ObsAirDis
    from &dsn1._Certs3
    where  CertCat = "OAD"
    order by patient_id;

/*PTSD*/
    create table &dsn1._PTSD_patient_ids as
    select distinct patient_id, 1 as PTSD
    from &dsn1._Certs3
    where  CertCat = "PTSD"
    order by patient_id;

/*Substance Abuse*/
    create table &dsn1._SA_patient_ids as
    select distinct patient_id, 1 as SubsAb
    from &dsn1._Certs3
    where  CertCat = "SA"
    order by patient_id;

/*Spinal*/
    create table &dsn1._Spinal_patient_ids as
    select distinct patient_id, 1 as Spinal
    from &dsn1._Certs3
    where  CertCat = "SP"
    order by patient_id;

/*Sarcoidosis*/
    create table &dsn1._Sarcoidosis_patient_ids as
    select distinct patient_id, 1 as Sarcoidosis
    from &dsn1._Certs3
    where  CertCat = "SRC"
    order by patient_id;

/*Upper Respiratory*/
    create table &dsn1._UpResp_patient_ids as
    select distinct patient_id, 1 as UpResp
    from &dsn1._Certs3
    where  CertCat = "URD"
    order by patient_id;

/*"ILD" "SRC"*/
    create table &dsn1._ILD_SRC_patient_ids as
    select distinct patient_id, 1 as ILDSRC
    from &dsn1._Certs3
    where  CertCat in ("ILD" "SRC")
```

```
    order by patient_id;

/*none*/
    create table &dsn1._No_Certs_patient_ids as
    select distinct patient_id, 1 as NoCerts
    from &dsn1._Certs3
    where  CertCat = "None"
    order by patient_id;

/*other*/
    create table &dsn1._Other_Certs_patient_ids as
    select distinct patient_id, 1 as Other_Certs
    from &dsn1._Certs3
    where  CertCat = "Other"
    order by patient_id;

quit;



/*merge all certification category counts with the main dataset*/

proc sql;
    create table &dsn1._sample as
    select distinct (patient_id) as patient_id
    from &dsn1._certs
    order by patient_id;
quit;

data &dsn1._Certifications;
    merge
    &dsn1._sample (in=a)
        &dsn1._cancer_patient_ids
        &dsn1._Adj_patient_ids
        &dsn1._Anx_patient_ids
        &dsn1._Dep_patient_ids
        &dsn1._Ext_patient_ids
        &dsn1._GERD_patient_ids
        &dsn1._Head_patient_ids
        &dsn1._ILD_patient_ids
        &dsn1._OAD_patient_ids
        &dsn1._PTSD_patient_ids
        &dsn1._SA_patient_ids
        &dsn1._Spinal_patient_ids
        &dsn1._Sarcoidosis_patient_ids
        &dsn1._UpResp_patient_ids
        &dsn1._ILD_SRC_patient_ids
        &dsn1._No_Certs_patient_ids
        &dsn1._Other_Certs_patient_ids
        ;
    by patient_id;
    if a;
    if Cancer = . then Cancer = 0;
    if Adjustment = . then Adjustment = 0;
    if Anxiety = . then Anxiety = 0;
    if Depression = . then Depression = 0;
    if Extremities = . then Extremities = 0;
    if GERD = . then GERD = 0;
    if Head = . then Head = 0;
    if IntLungDis = . then IntLungDis = 0;
    if ObsAirDis = . then ObsAirDis = 0;
    if PTSD = . then PTSD = 0;
    if SubsAb = . then SubsAb = 0;
    if Spinal = . then Spinal = 0;
    if Sarcoidosis = . then Sarcoidosis = 0;
    if UpResp = . then UpResp = 0;
    if ILDSRC = . then ILDSRC = 0;

    if max(of Cancer Adjustment Anxiety Depression Extremities GERD Head IntLungDis
    ObsAirDis PTSD SubsAb Spinal Sarcoidosis UpResp) ge 1 then AnyCert = 1; else AnyCert = 0;

    if max(of IntLungDis ObsAirDis Sarcoidosis) ge 1 then LowResp = 1; else LowResp = 0;

    if NoCerts = . then NoCerts = 0;
    if WTC1_Certs = . then WTC1_Certs = 0;
    if Other_Certs = . then Other_Certs = 0;
```

```
    if max(of Adjustment Anxiety Depression PTSD SubsAb) = 1 then MHCert = 1; else MHCert = 0;

    if max(of Extremities GERD Head IntLungDis ObsAirDis Spinal Sarcoidosis UpResp)= 1
    then PHCert = 1; else PHCert = 0;
run;


proc sql;
    title "counting total certifications per member";
    create table &dsn1._Certs4 as
    select distinct a.patient_id, max (AnyCert) as AnyCert, count(distinct CertCat) as TotCS,max (NoCerts) as NoCerts,
    max (MHCert) as MHCert,
    max (PHCert) as PHCert,
    max (Other_Certs) as Other_Certs,

    min(datepart(Start_Date)) as MinSDate format date9.,
    max(datepart(Start_Date)) as MaxSDate format date9.,
    max (Cancer) as cancer,
    max (Adjustment) as Adjustment,
    max (Anxiety) as Anxiety,
    max (Depression) as Depression,
    max (Extremities) as Extremities,
    max (GERD) as GERD,
    max (Head) as Head,
    max (IntLungDis) as IntLungDis,
    max (ObsAirDis) as ObsAirDis,
    max (PTSD) as PTSD,
    max (SubsAb) as SubsAb,
    max (Sarcoidosis) as Sarcoidosis,
    max (UpResp) as UpResp,
    max (ILDSRC) as ILDSRC,
    max (LowResp) as LowResp,
    max (WTC1_Certs) as WTC1_Certs

    from &dsn1._Certs3 a
    left join &dsn1._Certifications (keep = patient_id cancer Adjustment Anxiety
            Depression
            Extremities
            GERD
            Head
            IntLungDis
            ObsAirDis
            PTSD
            SubsAb
            Sarcoidosis
            UpResp
            ILDSRC
            AnyCert
            LowResp
            NoCerts
            WTC1_Certs
            Other_Certs
            MHCert
            PHCert) b
    on a.patient_id=b.patient_id
    group by a.patient_id
    order by a.patient_id;
quit;


data &dsn1._Cert_Summary ;
    set &dsn1._Certs4;
    if MinSDate = . then TotCS = 0;
run;

%mend certs ;

%certs (Frailty, Recruited_data);


/***************************************************************************
8- CREATE COMBINED DATASET WITH ALL REQUESTED DATA ELEMENTS
***************************************************************************/

proc sql;
    create table GRDC_data as
    select  *
```

```
    from demo_agegrps a
    left join recent_smoke b
        on a.patient_id=b.patient_id
    left join BMI_mean c
        on a.patient_id=c.patient_id
    left join Frailty_cert_summary d
        on a.patient_id=d.patient_id
    left join employment_pre911 e
        on a.patient_id=e.patient_id
    order by patient_id;
quit;


/*YOU CAN THEN MERGE THIS DATASET (USING PATIENT_ID) WITH THE DATASET GENERATED IN THE "1- FRAILTY PROJECT DOCUMENTATION - DA

data RECAP_and_GRDC;
    merge RECAP_data (in=a) GRDC_data;
    by patient_id;
    if a;
run;
```